

Section 11

Systems Programming ECE3502

Relocating loaders

Two Methods:

- 1) Modification record (for SIC/XE)
- 2) Relocation bit (for SIC)

Relocation in SIC:

- A **relocation bit** associated with each **word** of object code.
- The relocation bits are gathered together into a **bit mask** following the length indicator in each Text record.
- If bit=1, the corresponding word of **object code is relocated**.

Relocation in SIC

Line	Loc	Source statement			Object code	
5	0000	COPY	START	0		
10	0000	FIRST	STL	RETADR	140033	1
15	0003	CLOOP	JSUB	RDREC	481039	1
20	0006		LDA	LENGTH	000036	1
25	0009		COMP	ZERO	280030	1
30	000C		JEQ	ENDFIL	300015	1
35	000F		JSUB	WRREC	481061	1
40	0012		J	CLOOP	3C0003	1
45	0015	ENDFIL	LDA	EOF	00002A	1
50	0018		STA	BUFFER	0C0039	1
55	001B		LDA	THREE	00002D	1
60	001E		STA	LENGTH	0C0036	1
65	0021		JSUB	WRREC	481061	1
70	0024		LDL	RETADR	080033	1
75	0027		RSUB		4C0000	0
80	002A	EOF	BYTE	C'EOF'	454F46	0
85	002D	THREE	WORD	3	000003	0
90	0030	ZERO	WORD	0	000000	0
95	0033	RETADR	RESW	1		
100	0036	LENGTH	RESW	1		
105	0039	BUFFER	RESB	4096		

Relocation in SIC

```

110      .
115      .      SUBROUTINE TO READ RECORD INTO BUFFER
120      .
125      1039      RDREC      LDX      ZERO      040030      1
130      103C      LDA      ZERO      000030      1
135      103F      RLOOP      TD      INPUT      E0105D      1
140      1042      JEQ      RLOOP      30103F      1
145      1045      RD      INPUT      D8105D      1
150      1048      COMP      ZERO      280030      1
155      104B      JEQ      EXIT      301057      1
160      104E      STCH      BUFFER,X      548039      1
165      1051      TIX      MAXLEN      2C105E      1
170      1054      JLT      RLOOP      38103F      1
175      1057      EXIT      STX      LENGTH      100036      1
180      105A      RSUB      4C0000      0
185      105D      INPUT      BYTE      X'F1'      F1      0
190      105E      MAXLEN      WORD      4096      001000      0
195      .
200      .      SUBROUTINE TO WRITE RECORD FROM BUFFER
205      .
210      1061      WRREC      LDX      ZERO      040030      1
215      1064      WLOOP      TD      OUTPUT      E01079      1
220      1067      JEQ      WLOOP      301064      1
225      106A      LDCH      BUFFER,X      508039      1
230      106D      WD      OUTPUT      DC1079      1
235      1070      TIX      LENGTH      2C0036      1
240      1073      JLT      LOOP      381064      1
245      1076      RSUB      4C0000      0
250      1079      OUTPUT      BYTE      X'05'      05      0
255      END      FIRST

```

Relocation in SIC

H COPY	00000000107A	
T0000001E FFC1400334810390000362800303000154810613C000300002A0C003900002D		1111 1111 11
T00001E15E000C00364810610800334C0000454F46000003000000		1110 000
T0010391E FFC040030000030E0105D30103FD8105D2800303010575480392C105E38103F		1111 1111 11
T0010570A8001000364C0000F1001000		1000
T00106119FE0040030E01079301064508039DC10792C00363810644C000005		1111 1110 0
E000000		

You must cut the T record when the next object code would break the **3-byte alignment** of relocation bits.

One-Pass Assemblers

The main problem in trying to assemble a program in one pass involves **forward references**.

Why Do we need forward referencing?

- Data items
- Labels on instructions

Eliminate forward references:

- Data items are defined before they are referenced.
- But, forward references to labels on instructions cannot be eliminated as easily.
- Prohibit forward references to labels
 - Not applicable

One-Pass Assemblers

Loc.	Symbols	Instructions	reference	Obj. code
	COPY	START	1000	
	EOF	BYTE	C'EOF'	
	THREE	WORD	3	
	ZERO	WORD	0	
	RETADR	RESW	1	
	LENGTH	RESW	1	
	BUFFER	RESB	4096	
	FIRST	STL	RETADR	
	CLOOP	JSUB	RDREC	
		LDA	LENGTH	
		COMP	ZERO	
		JEQ	ENDFILL	
		JSUB	WRREC	
		J	CLOOP	
	ENDFILL	LDA	EOF	
		STA	BUFFER	
		LDA	THREE	
		STA	LENGTH	
		JSUB	WRREC	
		LDL	RETADR	

One-Pass Assemblers

Loc.	Symbols	Instructions	reference	Obj. code
	COPY	START	1000	
1000	EOF	BYTE	C'EOF'	454F46
1003	THREE	WORD	3	000003
1006	ZERO	WORD	0	000000
1009	RETADR	RESW	1	
100C	LENGTH	RESW	1	
100F	BUFFER	RESB	4096	
200F	FIRST	STL	RETADR	141009
2012	CLOOP	JSUB	RDREC	48****
		LDA	LENGTH	
		COMP	ZERO	
		JEQ	ENDFILL	
		JSUB	WRREC	
		J	CLOOP	
ENDFILL	LDA		EOF	
	STA		BUFFER	
	LDA		THREE	
	STA		LENGTH	
	JSUB		WRREC	
	LDL		RETADR	

Symbol	Value
EOF	1000
THREE	1003
ZERO	1006
RETADR	1009
LENGTH	100C
BUFFER	100F
FIRST	200F
CLOOP	2012
RDREC	

2013

Memory

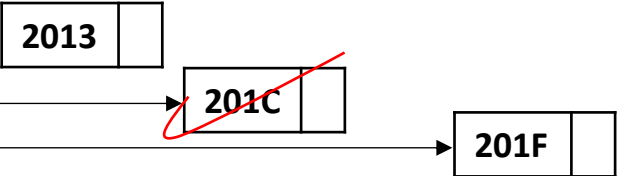
1000	45
1001	4F
1002	46
1003	00
1004	00
1005	03
1006	00
1007	00
1008	00

200F	14
2010	10
2011	09
2012	48
2013	**
2014	**

One-Pass Assemblers

Loc.	Symbols	Instructions	reference	Obj. code
	COPY	START	1000	
1000	EOF	BYTE	C'EOF'	454F46
1003	THREE	WORD	3	000003
1006	ZERO	WORD	0	000000
1009	RETADR	RESW	1	
100C	LENGTH	RESW	1	
100F	BUFFER	RESB	4096	
200F	FIRST	STL	RETADR	141009
2012	CLOOP	JSUB	RDREC	480000
2015		LDA	LENGTH	00100C
2018		COMP	ZERO	281006
201B		JEQ	ENDFILL	300000
201E		JSUB	WRREC	480000
2021		J	CLOOP	3C2012
2024	ENDFILL	LDA	EOF	001000
		STA	BUFFER	
		LDA	THREE	
		STA	LENGTH	
		JSUB	WRREC	
		LDL	RETADR	

Symbol	Value
EOF	1000
THREE	1003
ZERO	1006
RETADR	1009
LENGTH	100C
BUFFER	100F
FIRST	200F
CLOOP	2012
RDREC	
ENDFILL	2024
WRREC	



Memory

1000	45
1001	4F
1002	46
1003	00
1004	00
1005	03
1006	00
1007	00
1008	00

200F	14
2010	10
2011	09
2012	48
2013	**
2014	**
2015	00
2016	10
2017	0C
2018	28
2019	10
201A	06

201B	30
201C	20
201D	24
201E	48
201F	**
2020	**
2021	3C
2022	20
2023	12
2024	00
2025	10
2026	00

HTE record (Partial)

T 001000 09 454F46 000003 000000

T 00200F 15 141009 480000 00100C 281006 300000 480000 3C2012

T 00201C 02 2024

One-Pass Assemblers

Loc.	Symbols	Instructions	reference	Obj. code
	COPY	START	1000	
1000	EOF	BYTE	C'EOF'	454F46
1003	THREE	WORD	3	000003
1006	ZERO	WORD	0	000000
1009	RETADR	RESW	1	
100C	LENGTH	RESW	1	
100F	BUFFER	RESB	4096	
200F	FIRST	STL	RETADR	141009
2012	CLOOP	JSUB	RDREC	480000
2015		LDA	LENGTH	00100C
2018		COMP	ZERO	281006
201B		JEQ	ENDFILL	300000
201E		JSUB	WRREC	480000
2021		J	CLOOP	3C2012
2024	ENDFILL	LDA	EOF	001000
2027		STA	BUFFER	0C100F
202A		LDA	THREE	001003
202D		STA	LENGTH	0C100C
2030		JSUB	WRREC	480000
2033		LDL	RETADR	081009

Symbol	Value
EOF	1000
THREE	1003
ZERO	1006
RETADR	1009
LENGTH	100C
BUFFER	100F
FIRST	200F
CLOOP	2012
RDREC	203D
ENDFILL	2024
WRREC	2062

Memory

1000	45
1001	4F
1002	46
1003	00
1004	00
1005	03
1006	00
1007	00
1008	00

200F	14
2010	10
2011	09
2012	48
2013	20
2014	3D
2015	00
2016	10
2017	0C
2018	28
2019	10
201A	06

201B	30
201C	20
201D	24
201E	48
201F	20
2020	62
2021	3C
2022	20
2023	12
2024	00
2025	10
2026	00
2027	0C
2028	10
2029	0F

202A	00
201B	10
201C	03
201D	0C
201E	10
201F	0C

2030	48
2031	20
2032	62
2033	08
2034	10
2035	09

One-Pass Assemblers

Loc.	Symbols	Instructions	reference	Obj. code
2039	INPUT	BYTE	X'F1'	F1
203A	MAXLEN	WORD	4096	001000
203D	RDREC	LDX	ZERO	041006
2040		LDA	ZERO	001009
2043	RLOOP	TD	INPUT	E02039
2046		JEQ	RLOOP	302043
2049		RD	INPUT	D82039
204C		COMP	ZERO	281006
204F		JEQ	EXIT	300000
2052		STCH	BUFFER, X	54900F
2055		TIX	MAXLEN	2C203A
2058		JLT	RLOOP	382043
205B	EXIT	STX	LENGTH	10100C
205E		RSUB		4C0000
2061	OUTPUT	BYTE	X' 05'	05
2062	WRREC	LDX	ZERO	041006
2065	WLOOP	TD	OUTPUT	E02061
2068		JEQ	WLOOP	302065
206B		LDCH	BUFFER, X	50900F
206E		WD	OUTPUT	DC2061
2071		TIX	LENGTH	2C100C
2074		JLT	WLOOP	382065
2077		RSUB		4C0000
		END	FIRST	

Symbol	Value
EOF	1000
THREE	1003
ZERO	1006
RETADR	1009
LENGTH	100C
BUFFER	100F
FIRST	200F
CLOOP	2012
RDREC	203D
ENDFILL	2024
WRREC	2062
WLOOP	2065

Memory

1000	45
1001	4F
1002	46
1003	00
1004	00
1005	03
1006	00
1007	00
1008	00

200F	14
2010	10
2011	09
2012	48
2013	20
2014	3D
2015	00
2016	10
2017	0C
2018	28
2019	10
201A	06

201B	30
201C	20
201D	24
201E	48
201F	20
2020	62
2021	3C
2022	20
2023	12
2024	00
2025	10
2026	00
2027	0C
2028	10
2029	0F

202A	00
201B	10
201C	03
201D	0C
201E	10
201F	0C

2030	48
2031	20
2032	62
2033	08
2034	10
2035	09

HTE Record for One-Pass Assemblers

H COPY 001000 00107A

T 001000 09 454F46 000003 000000

T 00200F 15 141009 480000 00100C 281006 300000 480000 3C2012

T 00201C 02 2024

T 002024 19 001000 0C100F 001003 0C100C 480000 081009 4C0000 F1 001000

T 002013 02 203D

T 00203D 1E 041006 001006 E02039 302043 D82039 281006 300000 54900F 2C203A 382043

T 002050 02 205B

T 00205B 07 10100C 4C0000 05

T 00201F 02 2062

T 002031 02 2062

T 002062 18 041006 E02061 302065 50900F DC2061 2C100C 382065 4C0000

E 00200F